A natural boundary of dark matter haloes revealed around the minimum bias and maximum infall locations

Matthew Fong, Jiaxin Han

Speaker: Zhijun Zhang

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Introduction

- Dark matter haloes are the basic structure in the Universe
- The evolution of dark matter haloes determines galaxy evolutions and properties
- The understanding of what is the size of a halo is essential for studying dark matter halo evolution

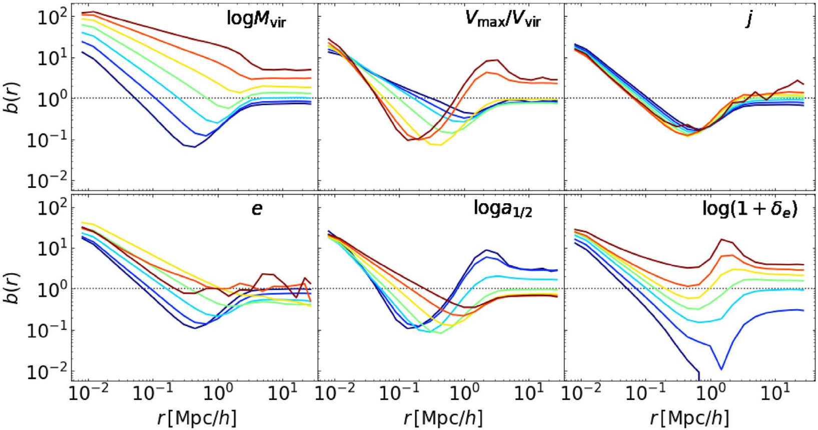
Dark Matter Halo Radius

- Virial Radius r_{vir} (spherical and static)
- Splashback Radius r_{sp} (density profile reaches its deepest slope)
- Turnaround Radius r_{ta} (Hubble flow velocity overcomes peculiar radial velocity)
- $r_{\text{vir}} \lesssim r_{\text{sp}} < r_{\text{ta}}$

Characteristic Depletion Radius

- CosmicGrowth Simulations (high resolution N body simulation with 3072^3 dark-matter particles)
- Bias profile (overdensity profile around a halo particle relative to average)
- Characteristic Depletion Radius r_{cd} (trough of the bias profile)
- Correlation between matter and haloes is the weakest
- Depends strongly on mass, environment and time, deeper trough = older

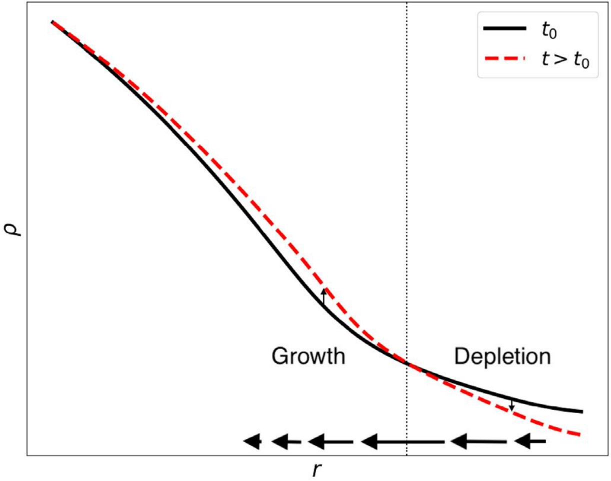
Characteristic Depletion Radius



Characteristic Depletion Radius

- 2-3 times the splashback radius
- 2.5 times the virial radius (mass independent)
- approximately the turnaround radius
- average density in r_{cd} is about 40 times the background

Characteristic Depletion Radius

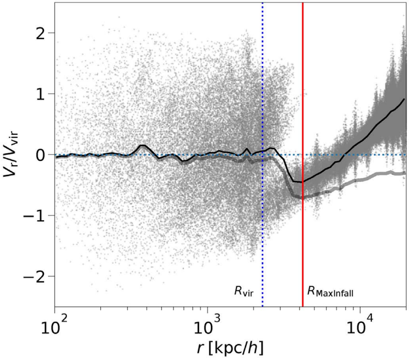


Inner Depletion Radius

- Inner Depletion Radius r_{id} (maximum mass inflow rate)
- Smaller than r_{cd} by $10 \sim 20\%$
- Matter drained from outside r_{id} leading the trough at r_{cd}

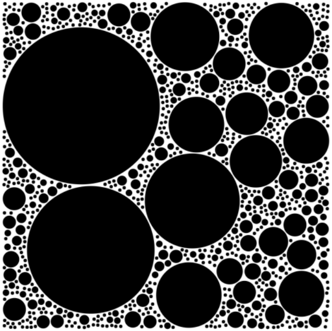
Characteristic Depletion Radius

- Natural boundary of splashback particles



Characteristic Depletion Radius

- Perfect radius of exclusion radius



Thank you for listening!